Synthesis, Welfare, and Student Research Designs

Behavioral Labor – Week 10

Teaching deck draft

Central question

- How do we translate behavioral labor ideas into welfare-relevant, empirically credible, frontier research projects?
- Week 10 is a capstone studio, not a review lecture.
- The goal is to move from a labor-market puzzle to a wedge, setting, design, method, welfare interpretation, and contribution.
- A good project integrates workers, workplaces, firms, markets, and policy design.

Why a final synthesis week matters

- Students can know many papers and still lack a research design.
- Behavioral labor claims are easy to overstate if treatment effects, mechanisms, welfare, and equilibrium are blurred.
- The field is strongest when behavioral objects are tied to labor-market margins and institutions.
- The final deliverable is a draftable research memo.

Course architecture in one map

Course block	Behavioral object	Research-design task
Weeks 1–4	preferences, beliefs, attention, learning	define the wedge and labor margin
Weeks 5–6	incentives, reciprocity, identity, norms	embed the wedge inside workplaces and social settings
Week 7	experiments, measurement, identification	separate treatment effects from behavioral objects
Weeks 8–9	firm response, markets, policy	add equilibrium, implementation, and welfare
Week 10	synthesis and project design	build a credible frontier question

What counts as a behavioral-labor contribution?

$$Y = g(B, I, M, P, X; \theta)$$

Object	Interpretation
B	behavioral wedge: beliefs, attention, present bias, fairness, identity, norms, learning
I	institution or environment: tax schedule, contract, benefit rule, workplace, platform
M	firm or market response: menus, monitoring, sorting, exploitation, accommodation
P	policy or program design: defaults, simplification, information, assistance, targeting
Y	labor outcome: work, effort, search, take-up, saving, training, sorting, earnings

A contribution needs all four pieces

- A real labor margin: search, effort, hours, saving, training, take-up, sorting, retention, promotion.
- A specific behavioral wedge, not a generic "friction."
- An institution where the wedge changes the labor prediction.
- A design that identifies the object being claimed.
- A bounded interpretation: treatment effect, behavioral parameter, welfare wedge, or equilibrium response.

Welfare and interpretation

$$W = \int \Psi_i\big(a_i, a_i^\star, M, P\big)\, dF(i)$$

- Observed choice may not fully reveal welfare when choices are distorted.
- The hard object is $a_i^\star$: full-information action, commitment-consistent action, long-run learned action, or model benchmark.
- Defaults, take-up, search beliefs, and attention interventions all require explicit normative criteria.
- A large behavioral response is not automatically a welfare gain.

Topic → wedge → setting

Topic family	Candidate wedge	Labor setting
Labor supply, saving, training	present bias, commitment, reference points	daily work, payroll saving, training enrollment
Job search and beliefs	biased expectations, slow learning	unemployment spells, applications, wage expectations
Attention and salience	perceived schedules, complexity	taxes, transfers, claiming, nonlinear contracts
Workplace behavior	reciprocity, fairness, monitoring response	effort, output, attendance, subjective evaluation
Identity, norms, culture	social image, norm compliance	sorting, retention, team behavior, promotion
Policy and firms	defaults, burden, targeting, market response	benefit design, menus, platforms, employer policy

Estimand versus interpretation

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0) \mid S_i = 1]$$

- Sometimes τ is exactly the object: did the intervention change the labor outcome?
- Sometimes the target is a behavioral parameter: belief bias, present bias, attention, reciprocity.
- Sometimes the target is welfare: corrected choice, targeting, or benchmark utility.
- Sometimes the target is equilibrium: firm response, spillovers, pass-through, or market adaptation.

Mapping setting to design and method

Setting	Common methods	Diagnostic question
Commitment or labor-supply menus	field experiments, treatment contrasts, structural recovery	does the design isolate self-control or schedule perception?
Repeated job-search beliefs	panel FE, forecast errors, hazards	do beliefs predict or respond to search outcomes?
Nonlinear schedules and knowledge	bunching, local elasticities, learning panels	statutory or perceived incentive?
Workplace reciprocity and monitoring	field experiments, heterogeneity, interactions	incentive response or social-preference channel?
Take-up and implementation	encouragement, hazard timing, admin data	claiming value or burden reduction?
Welfare and targeting	sufficient statistics, calibration, structure	what benchmark action is respected?

When reduced form is enough

- The project asks whether a clearly defined intervention changes a labor outcome.
- The behavioral channel is sharply embedded in treatment design.
- The contribution is local but meaningful: take-up, effort, search, training, saving, or claiming.
- Welfare and equilibrium claims are stated as limits or supported by auxiliary evidence.

When structure is needed

$$\hat{\psi} \in \arg\min_{\psi \in \Psi} \left[m^{data} - m^{model}(\psi)\right]' W \left[m^{data} - m^{model}(\psi)\right]$$

- The project needs unobserved parameters: present bias, learning, search costs, social preferences.
- The project makes counterfactual policy or targeting claims outside the treatment support.
- Welfare depends on a distorted-choice benchmark that is not directly observed.
- Equilibrium response or firm adaptation is central to interpretation.

Frontier project families

- Dynamic learning of policy schedules, eligibility, claiming, and training rules.
- Belief formation in rapidly changing labor markets.
- Algorithmic interfaces in search, scheduling, productivity, and benefits.
- Behavioral contracting, monitoring, feedback, and subjective evaluation inside firms.
- Behavioral heterogeneity, targeting, and distributional welfare.
- Firm-side exploitation, accommodation, insurance, and sorting around worker biases.

Why the frontier is still open

- Behavioral objects are hard to measure at scale.
- Worker response may mix beliefs, attention, constraints, trust, and stigma.
- Short-run treatment effects can decay, persist, diffuse, or reverse through learning.
- Firms and platforms adapt to predictable worker behavior.
- Welfare requires a normative benchmark, not just a coefficient.

Common failure modes

- Behavior is measured, but the behavioral wedge is not.
- The wedge is identified, but the labor-market margin is weak.
- A reduced-form effect is interpreted as a structural parameter.
- Learning dynamics are ignored.
- Firm or market response is offstage but central.
- A welfare claim is made without a benchmark.
- The project is interesting but not distinctly behavioral or not distinctly labor.

Research memo template

$$Contribution = Question + Mechanism + Data + Design + Interpretation$$

Memo component	What it must say
Question	labor-market puzzle or fact
Mechanism	behavioral wedge and standard alternative
Data	observed labor margin and behavioral object
Design	identifying variation, method, and estimand
Interpretation	welfare, policy, firm response, or equilibrium scope

Week 10 lab as proposal studio

- Reproduce: reverse-engineer one or two anchor papers' question and design logic.
- Diagnose: name the behavioral object, labor margin, method, welfare claim, and limit.
- Transfer: develop a mini-project memo in a new setting.
- Anchor menu: job-search beliefs, workplace social preferences, take-up frictions, retirement defaults.

Bridge outward

- A dissertation-quality project usually starts as a disciplined memo.
- The frontier rewards sharper objects, not broader lists of anomalies.
- The strongest projects connect worker behavior, firm response, policy design, and welfare.
- Behavioral Labor should leave students with open questions and research traction.

Takeaways

- Start from a labor margin and name the behavioral wedge.
- Embed the wedge in an institution, firm, market, or policy environment.
- Match the empirical method to the object being identified.
- Treat welfare and equilibrium as interpretation choices that need evidence.
- Turn synthesis into a research-design memo.